General Sir John Kotelawala Defence University



Faculty of Computing

Essentials of Artificial Intelligence

AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law

Group Number - 14	
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Table of Content

1. Abstract 3
2. Introduction 4
3. Problem Background 5
4. Problem statement.....6
5. Objectives 6
6. Selected AI techniques.....7
7. Justification of AI techniques.....8
8. System Design 9
9. Implementation Details 11
10. Testing and Results 12
11. Limitations and Discussion 14
12. Ethical and Social Considerations 15
13. Conclusion 16
14. References 17
15. Individual Contribution Sheet…..……19

1. Abstract

Legal research in common law systems specially in Sri Lanka depends on finding earlier judgments whose facts match with a current case. Doing this through printed law reports and keyword-based databases takes a massive time and effort. It often misses relevant cases that describe the same situation in different words. This project presents an AI-based case law recommendation system that takes the facts of a criminal case entered by a lawyer or law student and returns a ranked list of factually similar Sri Lankan criminal cases.

The system is built on an accurate dataset of approximately 1,000 verified Sri Lankan criminal cases from the New Law Reports (NLR) and Sri Lanka Law Reports , covering 1974 to 2015. Each record includes the case ID, case name, case facts and judgment. Case facts are cleaned and normalized using Natural Language Processing (NLP) techniques. The system then uses a hybrid retrieval approach. TF-IDF captures distinctive legal terms, section numbers and statutory references, while a Siamese Neural Network trained from scratch captures semantic similarity between case facts. Both scores are computed with cosine similarity and combined to rank the historical cases and the top K cases are returned. A TF-IDF only baseline was implemented to test whether the hybrid approach improves retrieval. Both were evaluated on Sri Lankan legal scenarios using Precision@1, Precision@5, Hit Rate@5 and Mean Reciprocal Rank . The results show that the hybrid approach retrieved relevant cases that TF-IDF only baseline missed where similar facts were described in different words while retaining TF-IDF's strength in matching exact legal terms and statutory references.. The system is delivered as a web application, and because it only returns cases from the verified dataset, it avoids fabricated case names, facts and judgments.

The system is meant for legal research purposes and not legal advice. Similarity scores reflect factual similarity, not legal applicability or predicted outcome. The report also covers the limitations of the work including dataset size, manual identification of similar case pairs and ethical considerations.

2. Introduction

Case law has an important place in the legal systems of common law countries, including Sri Lanka. Courts decide cases partly by looking at how earlier courts treated similar facts. So, lawyers prepare arguments and law students learn legal reasoning by studying past judgments. Therefore, finding the right earlier case is a core part of legal research.

The volume of reported decisions is massive and always keeps growing. Searching similar cases through it is time-consuming. Most available tools, such as the LawNet database and CommonLII rely on keyword search. This works when the user knows the exact terms used in a judgment but two cases can involve the same situation and be described in different languages. A researcher who does not guess the right words may miss relevant precedents entirely. At the same time, keywords still matter in law since section numbers and statutory references are highly distinctive. A useful system cannot ignore them.

Advances in NLP and neural networks make it possible to compare texts by meaning rather than exact wording. Text can be converted into numerical vectors called embeddings in which texts with similar meanings lie close together. However, general purpose generative AI tools can produce fabricated case names and judgments which is unacceptable in legal work. A legal research tool must therefore retrieve real cases from a trusted source.

This project develops an AI Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law. The user describes the facts of a case in plain language. The system preprocesses the text and compares it with a verified dataset of approximately 1,000 Sri Lankan criminal cases using a hybrid approach. TF-IDF weights distinctive legal terms and statutory references and a Siamese Neural Network trained from scratch on the project dataset measures semantic similarity. Cosine similarity is used to score the cases and the combined scores produce a ranked list. The system displays the case ID, name, key facts, judgment and similarity score of each recommended case. A TF-IDF only baseline is used to test whether the hybrid approach performs better than traditional keyword based retrieval.

The scope is limited to Sri Lankan criminal case law. The intended users are lawyers and law students. The system is a research tool. It does not give legal advice, decide whether a case applies or predict court outcomes.

3. Problem Background

Case law is the body of law formed by earlier court decisions. When lawyers handle a new matter they look for earlier judgments with comparable facts to see how courts reasoned and ruled.It helps them to build arguments and anticipate how a court may approach the issue. Law students study case law to understand how statutes are interpreted and applied in practice.

Traditionally, this research was done by hand through printed law reports such as the New Law Reports (NLR) and Sri Lanka Law Reports (SLR). Digital resources such as LawNet and CommonLII have made judgments easier to access but searching them still depends mainly on keywords.The researcher must read each candidate case to judge whether it is truly similar.

Keyword search has a basic weakness in legal research. Similar cases rarely use identical wording. One judgment may describe "possession of heroin concealed in luggage" while another discusses "knowledge in a drug importation charge". The facts are closely related but few words overlap. Finding the relevant case requires understanding similarity of meaning, not just matching terms. At the same time, purely semantic methods can overlook the exact section numbers and statutory references that lawyers rely on. Both kinds of relevance are needed.

Computational methods can help. NLP can process large volumes of legal text. Neural networks can learn representations that capture meaning. Prior work on legal case retrieval including document embeddings for identifying similar cases, legal knowledge graphs and contrastive learning approaches shows that semantic methods can improve the retrieval of related cases. However, most of this work addresses other jurisdictions such as the United States, India and Europe and little is available for Sri Lankan criminal law.

Reliability is a further concern. Legal professionals must cite real and verifiable authorities. But some AI tools generate plausible-looking but non existent cases. Therefore a suitable system should retrieve from a curated collection of real judgments instead of generating text.

This project addresses that gap by focusing on Sri Lankan criminal law and using a verified dataset of approximately 1,000 NLR and SLR cases from 1974 to 2015. It combines keyword based and semantic retrieval to support lawyers and law students in fact-based case retrieval.

4. Problem statement

Finding relevant Sri Lankan criminal case law for a given set of facts is difficult and time consuming. Lawyers and law students must review large numbers of past judgments. They need to choose effective search terms and recognise relevant cases themselves. Keyword based search which is the most common approach can miss cases with similar facts but described in different words. So important precedents may be overlooked.

Generative AI tools can speed up legal research. But they may produce fabricated case names, facts and judgments which makes them unsafe for legal work. In addition, there is no currently available tool designed to retrieve similar cases for Sri Lankan criminal law based on the similarity of case facts.

Therefore,this project aims to design, implement and evaluate such a system and to compare its hybrid TF-IDF and Siamese Neural Network approach against a TF-IDF only baseline.

5. Objectives

Main Objectives

The main goal is to create a web application that helps law students and legal professionals quickly find relevant past Sri Lankan criminal cases by simply typing in the facts of a situation.

Specific Objectives

Build a Legal Database: The task is to collect, clean and organize a database of historical Sri Lankan criminal court cases so they can be easily searched.,

Understand Case Meanings: The aim is to use Artificial Intelligence to understand the story and meaning behind a legal scenario rather than just searching for exact matching words.

Improve Search Accuracy: The goal is to combine search methods together so the system successfully finds cases that match both the facts of the crime and the specific laws broken.

Create a Simple Interface: The purpose is to design a user-friendly website that is easy for law students to use without requiring any advanced technical skills or complicated search commands.

6. Selected AI techniques

Natural Language Processing (NLP)

What it is: A branch of AI that helps computers understand, interpret and manipulate language.

Why it was selected: Court judgments contain a lot of text, abbreviations and long paragraphs. We used NLP techniques to clean the text remove unnecessary "stop words" (like and the is) and extract the core legal facts so the AI could read the cases properly.

TF-IDF (Term Frequency-Inverse Document Frequency)

What it is: A machine learning algorithm that calculates how important a specific word is to a document.

Why it was selected: In law specific words matter a lot. We selected TF-IDF because it is incredibly good at picking up statutory references (like "Section 296" or "Prevention of Terrorism Act") and ensuring cases with those exact laws are ranked at the top of the search results.

Siamese Neural Network (Deep Learning)
What it is: An neural network architecture that uses two identical AI models working side-by-side to compare two different inputs and see how similar they are.
Why it was selected: Sometimes people describe the crime using different words (e.g. "stolen vehicle" vs. "Car theft"). We selected a Siamese Neural Network because it understands the context and meaning of a sentence allowing the system to match cases based on the "story" of the crime even if the exact keywords don't match.

Cosine Similarity

What it is: A AI formula used to measure the angle between two points of data to see how closely they relate to each other.
Why it was selected: Once the Neural Network and TF-IDF algorithm finish reading the users search and the database cases we use Cosine Similarity to calculate a percentage score (e.g. "85% Match") to rank the most relevant cases, for the user.

7. Justification of AI techniques

Why use a Hybrid Approach (Two models of one)?

Think that using a Hybrid Approach is better than using one AI model. The Neural Network is very good at understanding the story. TF-IDF is very good at finding legal words and section numbers. By combining Neural Network and TF-IDF the system gets the best of both worlds. Does not miss anything.

Why Siamese Neural Networks?

Standard AI tries to sort things into categories like sorting pictures into cats or dogs. Siamese Neural Networks are different. They are built to compare two things. This is perfect for the app because the app directly compares the user’s typed story with cases to see how closely they match.

Why TF-IDF of normal search?

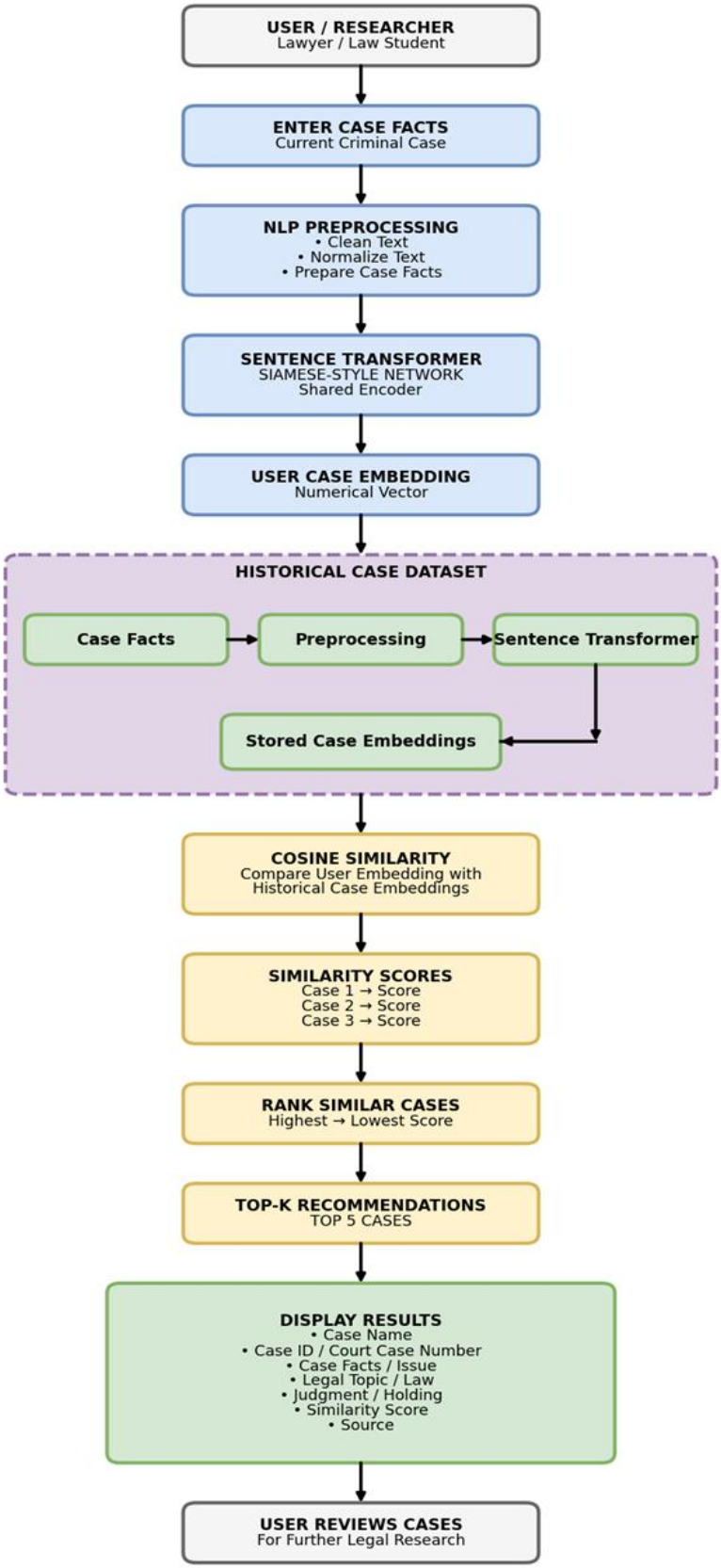
A normal search, like pressing Ctrl+F just looks for words and treats all words the same. TF-IDF is smarter. It automatically ignores useless words like the and court and focuses heavily on rare important legal words, like mens area or Section 296 to give better results.

Why build it from scratch?

Normal AI software often gets blocked by Windows security systems and antivirus programs. By programming the AI mathematics from scratch ourselves we made sure the app can run smoothly and safely on any computer without getting blocked.

8. System Design

The system provides an opportunity for attorneys and law students to input case information and then find similar cases in the criminal records of Sri Lanka. The system preprocesses the incoming text and then calculates the case embeddings by using a Siamese neural network with Sentence Transformer. Historical case embeddings are then compared to the user case using cosine similarity and the most similar cases are ranked. The top K cases are shown with the case detail, judgement/holding and a similarity score.


USER / RESEARCHER
Lawyer / Law Student
ENTER CASE FACTS
Current Criminal Case
NLP PREPROCESSING
• Clean Text
• Normalize Text
• Prepare Case Facts
SENTENCE TRANSFORMER
SIAMESE-STYLE NETWORK
Shared Encoder
USER CASE EMBEDDING
Numerical Vector
HISTORICAL CASE DATASET
Case Facts
Preprocessing
Sentence Transformer
Stored Case Embeddings
COSINE SIMILARITY
Compare User Embedding with
Historical Case Embeddings
SIMILARITY SCORES
Case 1 → Score
Case 2 → Score
Case 3 → Score
RANK SIMILAR CASES
Highest → Lowest Score
TOP-K RECOMMENDATIONS
TOP 5 CASES
DISPLAY RESULTS
• Case Name
• Case ID / Court Case Number
• Case Facts / Issue
• Legal Topic / Law
• Judgment / Holding
• Similarity Score
• Source
USER REVIEWS CASES
For Further Legal Research

9. Implementation Details

The system has been developed on the foundation of the NLP and machine learning techniques based on Python.

Data Processing

The Sri Lankan criminal case data set was cleaned and prepared for the AI process. Embeddings were produced after extracting and normalizing case facts.

Neural Network Implementation

The Siamese-style shared encoder was used, along with a pretrained Sentence Transformer. Embeddings are produced by the same encoder for the user case and historical case facts.Embeddings are created using the same encoder for the user case and historical case facts.

Similarity Calculation

Cosine similarity was used to compare the embeddings created. The higher the similarity score the better the case will be ranked on the recommendation list.

Case Recommendation

The system ranks the historical cases by similarity scores and shows the Top-K similar cases to the user.

Baseline Model

As a baseline we used a TF-IDF with Cosine Similarity model to compare traditional keyword-based retrieval with the approach of the neural network and the similarity of meaning.

User Interface

A prototype interface was created allowing users to enter case facts and see the cases recommended and their case information and similarity scores.

10. Testing and Results

Testing Approach

Various criminal case scenarios were used for testing the developed AI-based recommendation system for legal cases. A description of the facts of each test case was added to the system for each case. The system represented the input in terms of meaning and matched it with the historical Sri Lankan criminal cases data set. The cases were then sorted by their similarity score and the most relevant cases are shown to the user.

TestCase	Input Scenario	Recommended Case 1	Score	Recommended Case 2	Score	Recommended Case 3	Score	Status
TC 01	Murder / Culpable Homicide	Bandara vs	**54.9%**	100ALLES, Abcysundcni v. The Queen	**50.2%**	Saibu Vs. Attorney General	**48.4%**	Pass
TC 02	Bribery	J. K. H. M. H. Karunaratne, Appellant, and The Queen, Respondent	**46.1%**	J. Wijeratne, Appellant and The Republic of Sri Lanka, Respondent	**45.9%**	J. K. Nadaraja, Appellant and The Republic of Sri Lanka, Respondent	**45.9%**	Pass
TC 03	Theft	i2 Stephen v. Inspector of Police, Fort	**49.2%**	The Queen v. W. Wijepala	**48.3%**	164 Mariyathasan v. Rose	**46.2%**	Pass

TC 04	Assault / Grievous Hurt	Gomez Vs. State	**54.3%**	Bandara vs	**53.5%**	Dharmasena v. Sub-Inspector of Police, Kaduganna wa 339	**53.4%**	Pass
TC 05	Drug-related Offence	Hameed v. Inspector of Police, Crimes 9	**50.8%**	Weerasinghe v. Karunaratne 87	**42.3%**	296 T. S. Fernando, J. , The Attorney-General v. Karunaratne	**42.2%**	Pass

11. Limitations and Discussion

Even though the system worked and showed good results, it still has some limitations worth mentioning honestly.

The biggest one is the size of the dataset. We started with 770 Sri Lankan criminal cases and grew it to around 1,000 by the end of the project, but that is still small compared to the actual number of criminal cases in Sri Lanka. Rarer crime types are probably underrepresented, so the system likely performs better on common crimes than on rare ones.

Another problem is that there was no ready made dataset telling us which cases are similar and which are not. So our team had to go through cases manually and decide which pairs count as similar. This is a bit subjective, and a professional lawyer might not always agree with the pairs we chose.

We also built the AI part from scratch instead of using a big pretrained language model, mainly because pretrained AI libraries often get flagged or blocked by antivirus software, and we wanted the app to run smoothly on a normal computer. This helped usability, but it also means our model has seen far less text than a large pretrained model, limiting how well it understands unusually written case facts.

Lastly, the system covers only Sri Lankan criminal law, and its similarity score reflects how closely facts match , not legal correctness or case outcome. We also did not get to test it with real lawyers or law students outside the team, so real-world performance is still unconfirmed.

Overall, these limitations don't take away from what the project showed , that combining keyword matching with semantic understanding works better than either method alone , but they are honest points to improve if the project continues further.

12. Ethical and Social Considerations

Since this project deals with real court cases and is meant to be used by law students and lawyers, we had to think about a few ethical points while building it.

First, the system is only a research tool. It does not give legal advice, and it does not decide whether a case applies to another case legally. We were careful to keep this distinction clear so that no one mistakes the tool's output as an actual legal opinion.

Second, there is a risk that students could start depending on the tool too much instead of doing their own reading and analysis. Legal research is supposed to build critical thinking skills, so the app is meant to speed up the search process, not replace the thinking part of legal study.

Third, because some AI tools are known to make up fake case names or facts that sound real, we made sure our system can only recommend cases that actually exist in our database. It is not allowed to generate new case information on its own. This was an important decision because giving false legal information to a law student or lawyer could cause real problems.

We also used only publicly available court judgments for the dataset, which are already public documents, but we were still careful about how we stored and handled this data since judgments can mention private details about real people involved in the cases.

Another point is that since the dataset was built by us manually, there is a chance it does not represent all courts or time periods equally, which could cause the system to be biased toward certain types of cases. This is something that would need to be fixed with a more structured data collection process if the project is developed further.

On the positive side, if this system is improved and used properly, it could genuinely help reduce the time and effort law students and lawyers spend doing manual case research in Sri Lanka, especially for those who don't have easy access to expensive legal databases.

13. Conclusion

This project was about solving a problem that a lot of law students and lawyers in Sri Lanka actually face , spending too much time manually searching for similar past criminal cases, without getting stuck with the fake or made up results that some AI tools can produce. To solve this, we built a system that uses NLP to clean the case text, TF-IDF to catch important legal terms and section numbers, and a Siamese Neural Network to understand the overall story behind a case, and then used Cosine Similarity to rank and score the most relevant matches.

When we compared our hybrid system against a TF-IDF only baseline, it was clear that the hybrid approach found cases that pure keyword search missed, especially when the facts were described using different words. At the same time it still kept TF-IDF's strength of catching exact legal terms and statute references. Over the course of the project we also grew our dataset from 770 cases to around 1,000, and built a working interface where users can type in case facts in plain language and get back a ranked list of relevant past cases.

In the end, the system does what it set out to do , it helps users find factually similar Sri Lankan criminal cases faster, without pretending to give legal advice or predict outcomes. The limitations we found, like the dataset size and the lack of a proper labelled dataset, are not major flaws but realistic things to improve if this project is taken further after the module ends.

14. References

Abadi, M., Barham, P., Chen, J., Chen, Z., Davis, A., Dean, J., Devin, M., Ghemawat, S., Irving, G.,Isard, M., Kudlur, M., Levenberg, J., Monga, R., Moore, S., Murray, D. G., Steiner, B., Tucker, P., Vasudevan, V., Warden, P., et al. (2016). TensorFlow: Large-scale machine learning on heterogeneous systems Proceedings of the 12th USENIX Symposium on Operating Systems Design and Implementation (OSDI), 265–283.

Almuslim, I., & Inkpen, D. (2020). Document level embeddings for identifying similar legal cases and laws,in *Proceedings of the Forum for Information Retrieval Evaluation (FIRE 2020) Working Notes*, 42–48. https://ceur-ws.org/Vol-2826/T1-8.pdf

Bromley, J., Guyon, I., LeCun, Y., Säckinger, E., & Shah, R. (1994). Signature verification using a "Siamese" time delay neural network. *Advances in Neural Information Processing Systems (NeurIPS)*, 737–744.

CommonLII. (n.d.). *Sri Lanka case law databases*. Commonwealth Legal Information Institute. Retrieved August 19, 2026, from http://www.commonlii.org

Dasula, A. M., Tigulla, H., & Bhukya, P. (2024). Judgement citation retrieval using contextual similarity. *arXiv:2406.01609*. https://arxiv.org/abs/2406.01609

Dhani, J. S., Bhatt, R., Ganesan, B., Sirohi, P., & Bhatnagar, V. (2021). Similar cases recommendation using legal knowledge graphs. *arXiv:2107.04771*. https://arxiv.org/abs/2107.04771

Honnibal, M., & Montani, I. (2017). *spaCy 2: Natural language understanding with Bloom embeddings, convolutional neural networks and incremental parsing* [Computer software]. https://spacy.io

Neculoiu, P., Versteegh, M., & Rotaru, M. (2016). Learning text similarity with Siamese recurrent networks. *Proceedings of the 1st Workshop on Representation Learning for NLP (RepL4NLP)*, 148–157.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D.,
Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python.
Journal of Machine Learning Research, 12, 2825–2830.

Shu, D., Zhao, H., Liu, X., Demeter, D., Du, M., & Zhang, Y. (2024). LawLLM: Law large language model for the US legal system. *arXiv:2407.21065*. https://arxiv.org/abs/2407.21065

Sri Lanka Ministry of Justice. (n.d.). *LawNet: Sri Lanka legal database*. Retrieved August 19, 2026, from https://www.lawnet.gov.lk

Tang, Y., Qiu, R., Liu, Y., Li, X., & Huang, Z. (2024). LEXA: Legal case retrieval via graph contrastive learning with contextualised LLM embeddings. *arXiv:2405.11791*. https://arxiv.org/abs/2405.11791

Personal Contribution Report:

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INDIVIDUAL CONTRIBUTION SHEET

Student Name	**P.Dayawansha**	**Registration No.**	**D/COE/25/0014**
Degree / Intake	**Computer Engineering/42**	**Group No.**	**14**
Project Title	**AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law**		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
1. Legal Case Collection and Web Scraping	Identified reliable legal sources, including the New Law Reports and Sri Lanka Law Reports, and collected historical Sri Lankan criminal cases from LankaLaw. I used automated methods to make the collection process faster and more	Python, Requests, BeautifulSoup, Web Scraping	scraper_lankalaw.py sri_lanka_criminal_cases_corpus.csv Over 770 raw case records

2. Dataset Preparation and Case Categorization	Organized the 771 cases into seven legal categories, such as Penal Code offences, narcotics, and bail-related cases. This made the dataset easier to manage and helped prepare the legal information for the AI model.	NLP, Legal Case Categorization, Data Organization	Categorized cases in cleaned_criminal_cases.csv Dataset prepared for create_siamese_pairs.py
3. User Interface Design and Development	Contributed to designing the web application's interface for law students and legal professionals. I worked on its visual appearance, including the dark blue theme, information cards, and the layout used to display similar cases and their matching scores.	Streamlit, HTML, CSS	app.py Web interface with Similar Case Laws and Browse Database sections

4. Search History and Cloud Deployment	Contributed to implementing a feature that allowed users to revisit previous searches during their session. I also helped with the process of uploading the project to GitHub and making the application available online through Streamlit Community Cloud.	Streamlit Session State, Git, GitHub, Streamlit Community Cloud	https://github.com/Pabadi127/sri-lankan-legal-ai GitHub repository push_to_github.bat Deployed web application

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: _____ **Date:** 17/10/2026

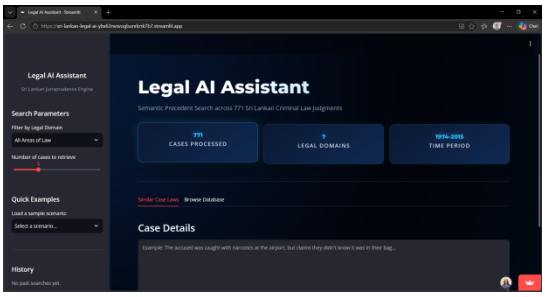
Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

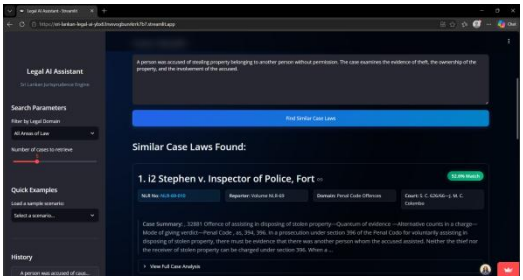
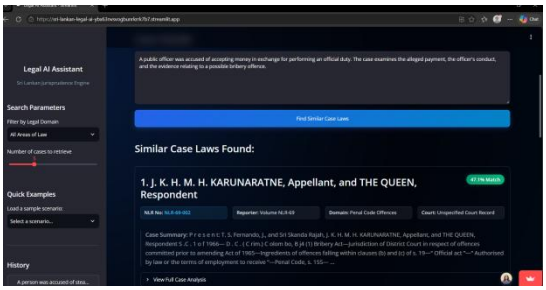
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INDIVIDUAL CONTRIBUTION SHEET

Student Name	W.S.T. Gunawardana	**Registration No.**	D/BIT/24/0052
Degree / Intake	BIT/41	**Group No.**	14
Project Title	AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law		

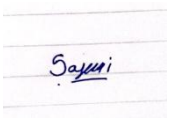
Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Frontend and backend interface	Developed the application interface for users to enter case facts and interact with the recommendation system.	Streamlit and Python	app.py 
AI model integration	Integrated the case-similarity model with the application so user-entered facts could be processed and	Sentence Transformer / Siamese-style encoder; cosine similarity	evaluation_comparison.csv

	recommendations returned.		
Case-law dataset integration and results display	Connected the Sri Lankan criminal case dataset to the application and displayed ranked recommendations with case names, facts, judgment/holding, court/year, and similarity scores.	Case-law dataset; similarity-ranking pipeline	 cleaned_criminal_cases.csv
System testing and refinement	Tested the application using different criminal case scenarios, checked the displayed recommendations and similarity scores, and refined the interface for usability.	Test scenarios; application testing	 evaluation_details.csv

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** __16.10.2026_____

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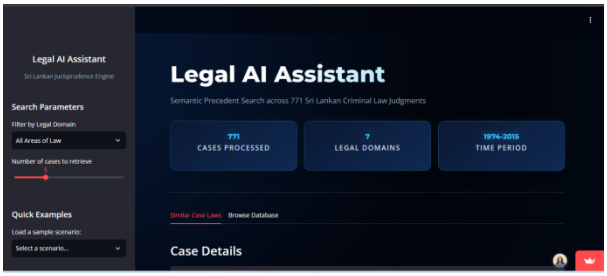
INDIVIDUAL CONTRIBUTION SHEET

Student Name	**K.S. Navodya**	**Registration No.**	**D/BIS/24/0040**
Degree / Intake	**BIS /41**	**Group No.**	**14**
Project Title	**AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law**		

Technical Contributions:

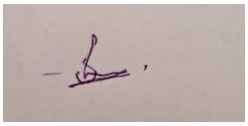
Task / Component	**My Contribution**	**Tools / AI Technique Used**	**Evidence / Output**

1. Siamese Neural Network Development (Main Responsibility)	Designed and implemented the twin-network projection architecture using dense hidden layers and developed the cosine similarity matching engine for semantic precedent retrieval.	Python, NumPy, Contrastive Learning, Cosine Similarity	Code: siamese_model.py Embeddings: siamese_case_embeddings.pkl GitHub commits (navo4)
2. Contrastive Pair Generation & Training Pipeline	Created automated generator for 3,000 contrastive training pairs and built training loop using contrastive loss and Adam optimizer.	Contrastive Loss , Adam Optimizer, Pair Generation	Code: create_siamese_pairs.py Code: train_siamese.py
3. Benchmarking	Built automated evaluation	Precision@K,	Code: evaluate_models.py

& Model Evaluation	framework to benchmark Siamese network against baseline TF-IDF model	Recall@K, Automated Benchmarking	CSV Reports: evaluation_comparison.csv
4. Support for Data Preparation & AI Processing	Assisted in cleaning, normalizing, and structuring raw legal text in primary dataset; optimized feature extraction and training hyper-parameters.	Python, Pandas, Text Preprocessing	Dataset: cleaned_criminal_cases.csv
5. Supporting Contribution to UI Development	Integrated backend Siamese inference engine into web interface for real-time query execution and precedent display.	Streamlit, Python, Web UI Integration	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: 

Date: ___16/10/2026_____

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.

ESSENTIALS OF ARTIFICIAL INTELLIGENCE

INDIVIDUAL CONTRIBUTION SHEET

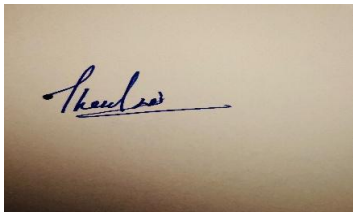
Student Name	**M. T. Jayaweera**	**Registration No.**	**D/DBA/25/0009**
Degree / Intake	**DBA / 42**	**Group No.**	**14**
Project Title	**AI-Based Legal Case Law Recommendation System Using Case Facts for Sri Lankan Criminal Law**		

Technical Contributions:

Task / Component	**My Contribution**	**Tools / AI Technique Used**	**Evidence / Output**
Text preprocessing	Built the pipeline that cleans and normalizes case-fact text (regex cleaning, OCR artefact removal, deduplication, stop word removal)	Python, NLP (spaCy / NLTK)	Preprocessing console output (case count, topic distribution), cleaned_criminal_cases.csv, cleaning script
TF-IDF baseline	Implemented the TF-IDF + Cosine Similarity retrieval baseline	scikit-learn (Tfidf Vectorizer, cosine similarity)	TF-IDF + Cosine Similarity code (notebook/script), TF-IDF row of the benchmark table
Benchmarking	Ran the baseline on the evaluation scenarios and compared it with the neural and hybrid results	Precision@1, Precision@5, Hit Rate@5, MRR	Evaluation table comparing TF-IDF, Siamese and hybrid on Precision@1, Precision@5, Hit Rate@5 and MRR
Re-run on expanded data	Refined preprocessing and re-ran the baseline for the larger dataset	Python	Updated evaluation results on the ~1,000-case dataset

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature:  **Date:** **17/10/2026**

Evidence may include source code, model/notebook, diagram, dataset work, test results, screenshots, report section or presentation slides.